



Cambridge (CIE) IGCSE Maths: Extended



Your notes

Proportion

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Direct Proportion



Your notes

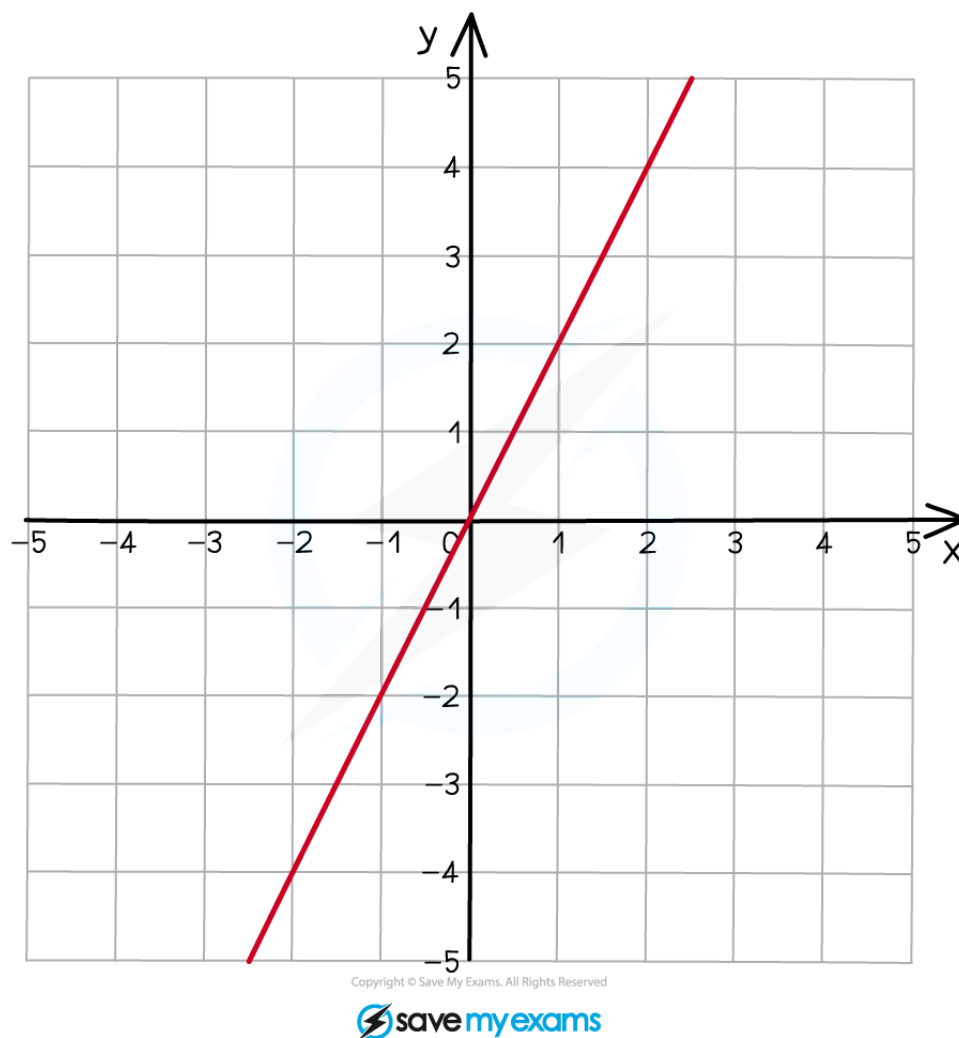
Direct Proportion

What is direct proportion?

- **Proportion** is a way of talking about how two **variables** are related to each other
- **Direct** proportion means that as one variable goes **up** the other goes up by the same **factor**
 - The **ratio** between the two amounts will always stay the **same**
- The symbol $\propto$ means "proportional to"
 - E.g. y is directly proportional to x , $y \propto x$
- If x and y are directly proportional, then
 - $x : y$ will always be the same
 - there will be some value, k , such that $y = kx$
 - the graph relating x and y is a linear graph, with **gradient k**
- k is called the **constant of proportionality**



Your notes



How do I use direct proportion with powers and roots?

- Problems may involve a variable being **directly proportional** to a **power or root** of another variable
- For example
 - y is directly proportional to the **square of x**
 - $y \propto x^2$
 - means that $y = kx^2$
 - y is directly proportional to the **square root of x**



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- $y \propto \sqrt{x}$
- means that $y = k\sqrt{x}$
- y is directly proportional to the **cube of x**
- $y \propto x^3$
- means that $y = kx^3$
- y is directly proportional to the **cube root of x**
- $y \propto \sqrt[3]{x}$
- means that $y = k\sqrt[3]{x}$
- Each of these would have a **different type of graph**, depending on the power or root

How do I find the equation between two directly proportional variables?

- Direct proportion questions always have the same process:
 - **STEP 1**
Identify the two variables and write down the **formula in terms of k**
 - E.g. y is **directly** proportional to x
 - write down the formula $y = kx$
 - **STEP 2**
Find k by substituting any given values **from the question** into your formula, then solving to get k
 - E.g. if you are told $y = 6$ when $x = 2$
 - then $6 = k \times 2$ giving $k = 3$
 - **STEP 3**
Rewrite the formula with the **known value of k** from above (substitute it in)
 - $y = 3x$
 - This is the **equation** relating the two variables
 - **STEP 4**
Use the **equation** to answer other parts of the question
 - E.g. find y when $x = 10$

- $y = 3x$ gives $y = 3 \times 10 = 30$



Examiner Tips and Tricks

- Some harder exam questions do not tell you to work out the equation
- You are expected to do it on your own



Worked Example

It is known that y is directly proportional to the square of x .

When $x = 3$, $y = 18$.

Find the value of y when $x = 4$.

Identify the two variables

$$y, x^2$$

We are told this is **direct** proportion

Write down the formula involving k

$$y = kx^2$$

Find k by substituting in $y = 18$ when $x = 3$

Then solve the equation for k

$$18 = k(3)^2$$

$$18 = 9k$$

$$\frac{18}{9} = k$$

$$2 = k$$

Substitute this value of k back into the formula to get the full equation

$$y = 2x^2$$

Use this formula to find y when $x = 4$



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$$y = 2 \times 4^2$$

$$y = 2 \times 16$$

$$y = 32$$

$$y = 32$$



Your notes



Your notes

Inverse Proportion

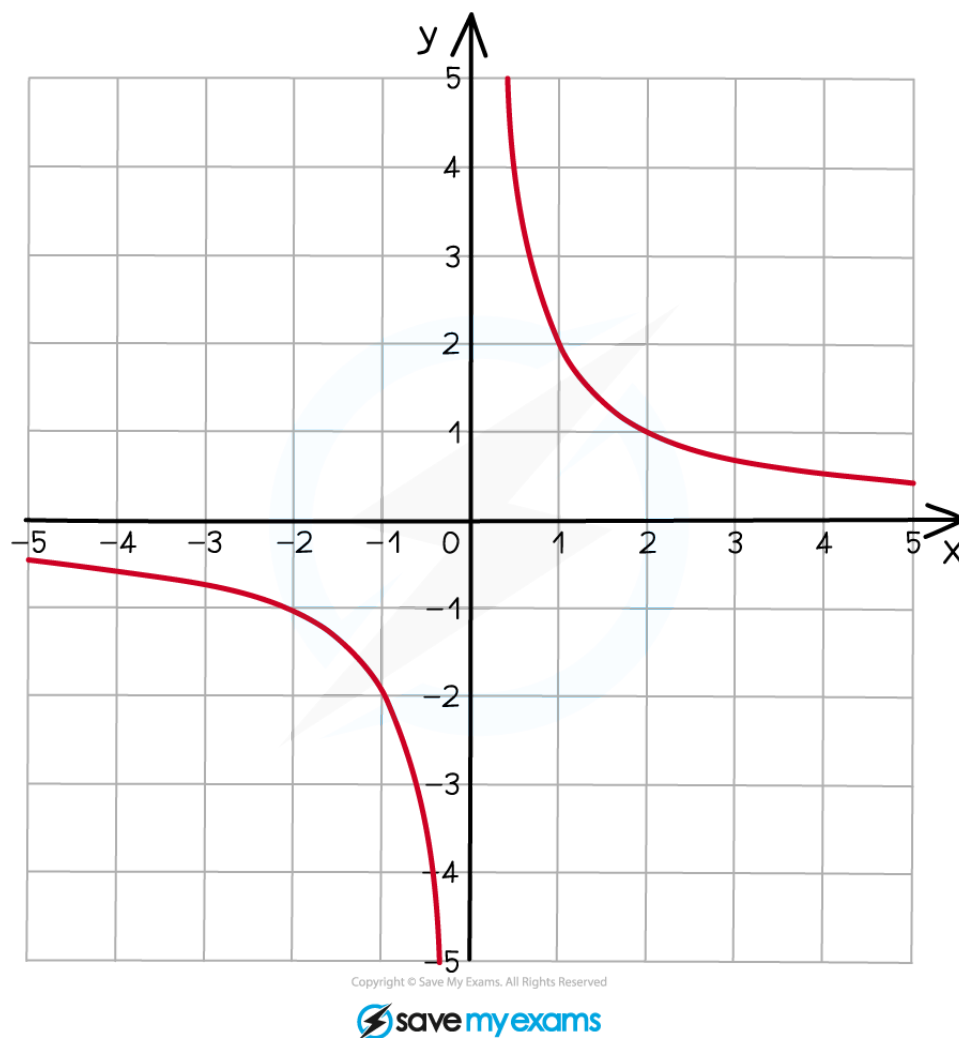
Inverse Proportion

What is inverse proportion?

- **Inverse** proportion means as **one variable goes up** the **other goes down** by the same **factor**
 - If two quantities are **inversely proportional**, then we can say that one is **directly proportional** to the **reciprocal** of the other
- The symbol, $\propto$, is used to show that one quantity is "directly proportional to the reciprocal of" (inversely proportional to) another quantity
 - "y is inversely proportional to x" is written $y \propto \frac{1}{x}$
- If x and y are inversely proportional then
 - $\frac{1}{x} : y$ will always be the same
 - there will be some value of k such that $y = \frac{k}{x}$
- The **graph** of $y = \frac{k}{x}$ is shown below



Your notes



How do I use inverse proportion with powers and roots?

- Problems may involve a variable being **inversely proportional** to a **power or root** of another variable
- For example
 - y is **inversely** proportional to the **square of x**

- $y \propto \frac{1}{x^2}$



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- means that $y = \frac{k}{x^2}$
- y is **inversely** proportional to the **square root of x**
 - $y \propto \frac{1}{\sqrt{x}}$
 - means that $y = \frac{k}{\sqrt{x}}$
- y is **inversely** proportional to the **cube of x**
 - $y \propto \frac{1}{x^3}$
 - means that $y = \frac{k}{x^3}$
- y is **inversely** proportional to the **cube root of x**
 - $y \propto \frac{1}{\sqrt[3]{x}}$
 - means that $y = \frac{k}{\sqrt[3]{x}}$
- Each of these would have a **different type of graph**, depending on the power or root

How do I find the equation between two inversely proportional variables?

- Inverse proportion questions always have the same process:
 - **STEP 1**
Identify the two variables and write down the **formula in terms of k**
 - E.g. y is **inversely** proportional to x
 - write down the formula $y = \frac{k}{x}$
 - **STEP 2**
Find k by substituting any given values **from the question** into your formula, then solving to get k



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- E.g. if you are told $y = 5$ when $x = 6$

- then $5 = \frac{k}{6}$ giving $k = 30$

- **STEP 3**

Rewrite the formula with the **known value of k** from above (substitute it in)

- $y = \frac{30}{x}$

- This is the **equation** relating the two variables

- **STEP 4**

Use the **equation** to answer other parts of the question

- E.g. find y when $x = 2$

- $y = \frac{30}{x}$ gives $y = \frac{30}{2} = 15$



Worked Example

The time, t hours, it takes to complete a project is inversely proportional to the cube root of the number, n , of people working on it.

If 27 people work on the project, it takes 50 hours to complete.

(a) Find an equation connecting t and n .

Identify the two variables

$$t, n$$

We are told this is **inverse** proportion to the cube root of n

Write down the formula involving k

$$t = \frac{k}{\sqrt[3]{n}}$$

Find k by substituting in $n = 27$ and $t = 50$ (from the words in the question)



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$$50 = \frac{k}{\sqrt[3]{27}}$$

$$50 = \frac{k}{3}$$

$$50 \times 3 = k$$

$$150 = k$$

Substitute this value of k back into the formula to get the full equation

$$t = \frac{150}{\sqrt[3]{n}}$$

(b) Given that the project needs to be completed within 60 hours, find the minimum number of people needed to work on it.

Use the formula to find n when $t = 60$

$$60 = \frac{150}{\sqrt[3]{n}}$$

$$\sqrt[3]{n} = \frac{150}{60} = 2.5$$

$$n = 2.5^3 = 15.625$$

A sensible answer here is a whole number (as n is the number of people)
15 people would not complete it in time, but 16 would

16 people